18 · Results, Part 2: What the Experiments Show

Step 1: The main result

	Recall@10	MRR@10	nDCG@10	Latency (p50)
Baseline (LLM → text → embedder)	0.637	0.329	0.402	43.5 ms
Projection head	0.607	0.293	0.367	2.0 ms
Difference	−3.0 pts	−3.6 pts	−3.5 pts	21.8× faster

Reading it: the head is a little worse at search and much faster at the encode step. The gap is small but statistically real ($p = 0.0005$).

A note on "97%": the −3.0 is in **percentage points**. As a ratio, the head keeps 0.607 / 0.637 = **95.3%** of the baseline's Recall@10, 89% of its MRR and 91% of its nDCG. The paper's "97%" and "95–97%" figures read like 100 − 3 rather than a ratio. See 19.

Step 2: The loss ablation (which losses matter)

All at 30 epochs, lr 3×10^{-4}:

Losses used	R@10	Lesson
Rank only	0.001	Collapses: no anchor in space

Losses used	R@10	Lesson
Contrastive only	0.498	Works, but weak alone
Align only	0.567	The strongest single signal
Align + Rank	0.579	Rank helps once alignment anchors things
Align + Contrastive	0.582	Separation helps
All three	0.595	Best combination
All three + long/slow recipe	**0.607**	Best overall

Step 3: The training recipe (how to train)

- Learning rate 5×10^{-4} collapses; 2×10^{-4} with 80 epochs is best.
- 50 epochs at 3×10^{-4} is **worse** than 30 epochs at the same rate, a sign of overshooting or overfitting when the rate isn't lowered.
- Going from the standard to the extended recipe (+0.012) gave the paper's largest single gain, larger than adding any one loss on top of the others.

Step 4: Loss-weight sensitivity

With no contrastive loss ($\lambda_c = 0$):

Setting	R@10
More alignment ($\lambda_a = 1.0, \lambda_r = 0.5$)	0.591
More rank ($\lambda_a = 0.5, \lambda_r = 1.0$)	0.516
Top-256 instead of top-128 docs	0.584

Leaning on alignment is safe; leaning on rank hurts. Doubling K doesn't help, so the ranking signal saturates at a moderate number of candidates.

Step 5: Where it fails

- Per-conversation agreement with the baseline averages 84.5%, but failures **cluster**: 48 conversations (with at least 3 triggers each) have recall well below the baseline.
- The paper **hypothesizes** (it doesn't test) that failures involve rare terms, long coreference chains, or domain vocabulary, the "long tail" where the head saw few training examples.
- It also points to the **causal vs. bidirectional** mismatch (file `04`): the head must translate from states that only look backward into a space built by a model that looks both ways.

Step 6: The takeaway

Claim	Supported?
A small head can stand in for the embedder at search time	Yes: 95% of recall, same index
It is much faster at the encode step	Yes: 43.5 → 2.0 ms
All three losses contribute	Yes, in this setting
Careful optimization matters a lot	Yes
Hidden states make search *better* than text (Problem 3)	Not shown: the head is slightly worse than the text pipeline